

Gonadotropin-Releasing Hormone (GnRH) Agonists

ACG: A-0304 (AC)

[Link to Codes](#)

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Clinical Indications

- Gonadotropin-releasing hormone (GnRH) agonists may be indicated for **1 or more** of the following:
 - Assisted reproductive technology (eg, female patient undergoing in vitro fertilization)(19)(20)(21)(22)(23)[N](#)
 - ☐ Cancer, as indicated by **1 or more** of the following:
 - Breast cancer, with **1 or more** of the following(29)(30):[N](#)
 - Adjuvant therapy needed, and **ALL** of the following(32)(34)(45):
 - Administered in combination with tamoxifen or an aromatase inhibitor (eg, exemestane)
 - Patient is premenopausal or male.
 - Tumor is estrogen receptor positive or progesterone receptor positive.
 - Advanced disease, and **ALL** of the following(1):
 - Palliative treatment
 - Patient is premenopausal or perimenopausal.
 - Prevention of premature ovarian failure needed, and **ALL** of the following(36)(40)(41)(42)(43):
 - Patient is receiving cytotoxic agent associated with premature ovarian failure (eg, cyclophosphamide).
 - Patient is premenopausal.
 - Prostate cancer, as indicated by **1 or more** of the following(1)(46)(47)(48)(49)(50):[N](#)
 - Intermediate-risk, high-risk, or very high-risk disease, as indicated by **1 or more** of the following(57)(58)(59)(60):
 - International Society of Urological Pathology (ISUP) Grade Group 2 to 5 (Gleason score of 7 to 10)(61)
 - Pretreatment PSA of 10 ng/mL (mcg/L) or greater
 - Stage T2b/T2c, stage T3a/T3b, or stage 4 prostate cancer
 - Metastatic prostate cancer (ie, bone or other metastasis)(62)(63)(64)(65)(66)(67)
 - Post radical prostatectomy with adverse laboratory, histologic, or biopsy features, as indicated by **1 or more** of the following(47):
 - Extracapsular extension of tumor
 - Invasion of seminal vesicle
 - Lymph node metastasis
 - Positive biopsy margin(s)
 - PSA detectable

- ☐ Central precocious puberty, as indicated by **1 or more** of the following^{[A](2)(5)(68)(71)(74)(77)(78):N}
 - Initial course, as indicated by **ALL** of the following:
 - Advanced bone age or accelerated growth velocity^{[B](89)}
 - Clinical signs or symptoms of precocious puberty, as indicated by **1 or more** of the following(90)(91)(92):
 - Female with **ALL** of the following:
 - Age between 2 and 8 years when signs or symptoms of precocious puberty appear
 - Tanner stage 2 or greater clinical findings (ie, breast and pubic hair development)
 - No pregnancy
 - No undiagnosed abnormal vaginal bleeding
 - Male with **ALL** of the following:
 - Age between 2 and 9 years when signs or symptoms of precocious puberty appear
 - Testicular volume 4 mL or greater
 - Laboratory confirmation of diagnosis, as indicated by **1 or more** of the following(89)(91)(93):
 - Pubertal basal level of luteinizing hormone (based on pediatric reference ranges)
 - Pubertal basal level of sex hormones (based on pediatric reference ranges)
 - Pubertal response of luteinizing hormone or sex hormone to stimulation by GnRH
 - Other causes of precocious puberty ruled out by appropriate hormonal and imaging studies (eg, congenital adrenal hyperplasia, CNS tumor)
 - Subsequent course, with favorable response to prior administration of GnRH agonist^{[C](89)}
- ☐ Dysfunctional uterine bleeding, as indicated by **ALL** of the following^{[D](1)(95):N}
 - Prior to planned endometrial ablation for definitive treatment
 - No current breast-feeding
 - No pregnancy currently or anticipated while receiving medication
 - Other causes of symptoms or bleeding ruled out (eg, by endometrial biopsy)
- ☐ Endometriosis, as indicated by **1 or more** of the following^{[E](1)(97)(98)(99)(100)(101):N}
 - Initial course, as indicated by **ALL** of the following:
 - Age 18 years or older
 - Endometriosis symptoms, as indicated by **1 or more** of the following(104):
 - Dysmenorrhea
 - Dyspareunia
 - Pelvic pain
 - Failure of or contraindication to **ALL** of the following(105):
 - Oral contraceptive pills or medroxyprogesterone acetate injection
 - Nonsteroidal anti-inflammatory drugs (NSAIDs)
 - No current breast-feeding
 - No pregnancy currently or anticipated while receiving medication
 - No vaginal bleeding of unknown cause
 - Subsequent course, as indicate by **ALL** of the following:
 - Age 18 years or older
 - Concurrent use of medication to counter anti-estrogen effects of GnRH agonist (eg, norethindrone acetate or bisphosphonate add-back therapy), unless contraindicated(104)(105)(106)(107)(108)
 - Patient has not received more than one previous 6-month course of GnRH agonist for treatment of endometriosis.
 - Recurrence of endometriosis symptoms following initial course of therapy
- ☐ Gender incongruence, as indicated by **1 or more** of the following^{[F](109)(110)(111)(112):N}
 - Initial course, as indicated by **ALL** of the following:
 - Marked and sustained gender incongruence^[G] as assessed and documented by clinician experienced in care of transgender and gender-diverse people
 - Pubertal development of Tanner stage 2 or greater
 - Subsequent course, as indicated by **ALL** of the following:
 - Marked and sustained gender incongruence^[G] as assessed and documented by clinician experienced in care of transgender and gender-diverse people
 - Favorable response to initial course of GnRH agonist
- ☐ Uterine leiomyomas (fibroids), as indicated by **ALL** of the following^{[H](97)(98)(121):N}
 - Age 18 years or older
 - Goal to reduce leiomyoma (fibroid) size or bleeding prior to operative intervention(124)
 - Leiomyoma symptoms, as indicated by **1 or more** of the following:
 - Abnormal uterine bleeding
 - Bulk-related symptoms (eg, pelvic pain or pressure, dyspareunia, urinary symptoms)
 - Iron deficiency anemia
 - No current breast-feeding
 - No pregnancy currently or anticipated while receiving medication

- Other causes of symptoms or bleeding ruled out (eg, by endometrial biopsy)

Evidence Summary

Background

GnRH agonists initially stimulate the release of the pituitary gonadotropins LH and FSH, resulting in a temporary increase in gonadal steroidogenesis, followed after repeated administration by longer-term abolition of the stimulatory effects of the pituitary gland.(1) **(EG 2)** This leads to decreased gonadal steroidogenesis, with reduction in functionality of tissues and processes dependent upon such steroids.(1)(2) **(EG 2)** Side effects are often significant, ranging from effects on libido and sexual function to increased risk of metabolic syndrome, osteoporosis and associated fractures, diabetes, and cardiovascular disease.(3) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 84 published peer reviewed articles, 18 specialty society or other evidence-based guidelines, 8 Cochrane systematic reviews, and 2 book sections.

For assisted reproductive technology, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Systematic reviews reported that when long (ie, for at least 14 days prior to ovarian stimulation) and short GnRH agonist protocols were compared, there was no conclusive evidence of differences in live birth and ongoing pregnancy rates; however, there was moderate-quality evidence of higher clinical pregnancy rates in the long-protocol group.(24)(25) **(EG 2)** Another systematic review and meta-analysis indicated that GnRH agonists were successful in significantly improving pregnancy rates when administered for 3 months in women with endometriosis who were undergoing assisted reproduction.(26)(27) **(EG 1)** An open-label randomized trial of 1344 patients undergoing in vitro fertilization compared luteal progesterone with luteal triptorelin (given as multiple injections or a single bolus injection) and found that triptorelin was associated with increased clinical pregnancy and delivery rates as compared with progesterone.(28) **(EG 1)**

For breast cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A randomized controlled trial randomized 3066 premenopausal women with hormone receptor-positive breast cancer to tamoxifen alone, tamoxifen plus ovarian suppression, or exemestane plus ovarian suppression for 5 years; ovarian suppression was achieved by either triptorelin intramuscular injection every 28 days, bilateral oophorectomy, or bilateral ovarian irradiation. There was no significant difference in disease-free survival between tamoxifen alone and tamoxifen plus ovarian suppression at 67-month follow-up. However, a subgroup analysis of 1628 women who had received prior chemotherapy found that ovarian suppression was associated with improved rates of freedom from breast cancer at 5 years (78% with tamoxifen alone, 82.5% with tamoxifen and ovarian suppression, and 85.7% with exemestane and ovarian suppression).(31) **(EG 1)** Extensions of this study found, at 8 years and 12 years of follow-up, that tamoxifen combined with ovarian suppression was associated with higher rates of disease-free and overall survival compared with tamoxifen alone.(32)(33) **(EG 1)** A specialty society guideline recommends ovarian suppression in addition to adjuvant endocrine therapy for higher-risk premenopausal patients with estrogen receptor-positive tumors; however, risk is not clearly defined.(34) **(EG 2)** Expert consensus guidelines recommend tamoxifen, either with or without ovarian suppression, as adjuvant therapy for premenopausal women with lymph node-positive disease; treatment with an aromatase inhibitor plus ovarian suppression is an alternate option for premenopausal women who are at increased risk of recurrence due to young age, high-grade tumor, or lymph node involvement. Guidelines recommend ovarian suppression with a luteinizing hormone-releasing hormone agonist for premenopausal women with metastatic disease.(29) **(EG 2)** A systematic review and meta-analysis evaluating the use of GnRH agonists for gonadal protection in women undergoing gonadotoxic chemotherapy included 10 studies of breast cancer and found that GnRH agonists were associated with increased rates of return of spontaneous menstruation as compared with controls.(35) **(EG 1)** A systematic review evaluating the use of GnRH agonists for the prevention of chemotherapy-induced ovarian failure (defined as a combination of amenorrhea and postmenopausal FSH levels) in premenopausal women that included 4 studies (780 patients with breast cancer) comparing treatment with chemotherapy alone or combined with GnRH agonists found moderate-quality evidence that patients treated with GnRH agonists plus chemotherapy had a lower incidence of chemotherapy-related premature ovarian failure compared with patients treated with chemotherapy alone (10.7% and 25.3%, respectively).(36) **(EG 1)** A meta-analysis of individual patient-level data from 873 premenopausal participants with a diagnosis of breast cancer evaluated the use of GnRH agonists during chemotherapy for the preservation of ovarian function and fertility and found that treatment with GnRH agonists was associated with a lower incidence of chemotherapy-induced premature ovarian insufficiency and amenorrhea and a higher incidence of post-treatment pregnancy compared with chemotherapy alone; no difference in the incidence of disease-free survival or overall survival was noted between the 2 treatment groups.(37) **(EG 1)** Long-term follow-up of an open-label phase III randomized trial of 281 premenopausal patients with breast cancer evaluating the safety of GnRH agonist therapy during chemotherapy for ovarian protection found, at a mean follow-up of 12.4 years, no difference in disease-free or overall survival in the GnRH agonist plus chemotherapy group compared with the chemotherapy-only group.(38) **(EG 1)** Another long-term follow-up of a randomized trial of 218 premenopausal patients with stage I to IIIA hormone receptor-negative breast cancer treated with either chemotherapy plus GnRH agonists or chemotherapy alone found, at a mean follow-up of 5.1 years, that more patients treated with chemotherapy plus GnRH agonists reported at least one post-treatment pregnancy compared with patients treated with chemotherapy alone. There was no statistical difference in disease-free survival or overall survival between the treatment groups.(39) **(EG 2)** An expert consensus guideline states that, regardless of hormone receptor status, ovarian suppression with GnRH agonists administered with adjuvant chemotherapy in premenopausal patients with breast cancers may preserve ovarian function and reduce the likelihood of chemotherapy-induced amenorrhea.(30) **(EG 2)** Multiple specialty society practice guidelines recommend the use of GnRH agonists

during chemotherapy in premenopausal patients with breast cancer as an option for ovarian function protection; however, the evidence is insufficient to recommend the use of GnRH agonists as a replacement for standard fertility preservation methods (ie, cryopreservation techniques).(40)(41)(42)(43) **(EG 2)** An expert consensus guideline on male breast cancer management states that GnRH agonists combined with aromatase inhibitors are an option for adjuvant treatment of men with hormone receptor-positive cancer who are not candidates for tamoxifen, and for men with hormone receptor-positive, HER2-negative cancer without rapidly progressive disease or visceral crisis.(44) **(EG 2)**

For prostate cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Studies indicate effectiveness of GnRH agonists in substantially reducing serum testosterone levels for palliation of signs and symptoms of advanced cancer.(51)(52)(53)(54) **(EG 2)** A meta-analysis of 16 studies of different GnRH agonists used for prostate cancer treatment concluded that there is insufficient evidence to directly compare efficacy and safety between these medications; further studies are recommended to determine comparative survival and toxicity among these treatments.(55) **(EG 1)** Specialty society guidelines recommend the use of GnRH agonists for certain patients with intermediate-risk, high-risk, or very high-risk disease, and for those with adverse findings after radical prostatectomy, castration-resistant disease, or metastatic disease.(47)(56) **(EG 2)**

For central precocious puberty, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty guideline consortium statement and review articles have concluded that GnRH agonists should be considered for children with progression of pubertal development in the setting of confirmatory diagnostic evaluation, including assessment of bone age and growth velocity, hormone levels, hormone stimulation testing, and central nervous system imaging.(79)(80)(81) **(EG 2)** A meta-analysis identified 8 studies of girls with precocious puberty and determined that treatment with GnRH agonists may have a positive effect on final adult height, but the results were not conclusive.(82) **(EG 1)** A prospective open-label study of 36 children with central precocious puberty suggested that histrelin implant may significantly improve predicted adult height in girls.(73) **(EG 2)** A meta-analysis of 5 studies involving 153 children with central precocious puberty found that triptorelin was effective in significantly suppressing the luteinizing hormone peak and other gonadal hormones and in slowing disease progression.(83) **(EG 1)** A phase III single-arm trial of 62 untreated pediatric patients with central precocious puberty treated with subcutaneous leuprolide every 6 months found, at 24-week follow-up, that 87% of patients had suppression of peak luteinizing hormone levels, and 57.4% had a decrease in growth velocity.(84) **(EG 2)** An open-label noncomparative phase III trial of 44 children with central precocious puberty found that the 6-month formulation of triptorelin suppressed serum luteinizing hormone levels in 97.7% of patients at 12 months, and the Tanner stage was stable or reduced in 88.6% at 12 months.(85) **(EG 2)** An open-label noncomparative phase III trial of 45 children with central precocious puberty found that the 6-month formulation of leuprolide acetate depot suppressed luteinizing hormone levels in 86.7% of patients at 6 months. Physical signs of puberty were suppressed at week 48 in 90.2% of girls and 75.0% of boys.(86) **(EG 2)** A review article states that children with central precocious puberty who are treated with GnRH agonists show stabilization or regression of physical signs of puberty and exhibit slowing of height velocity and growth plate maturation, without adverse effects on future fertility.(78) **(EG 2)**

For dysfunctional uterine bleeding, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** GnRH agonists are indicated for endometrial thinning prior to endometrial ablation.(1) **(EG 2)** A systematic review and meta-analysis found evidence supporting the effectiveness of GnRH agonists, as compared with no treatment, in significantly reducing postoperative dysmenorrhea as well as bleeding for up to 24 months after ablation, although the authors indicated that most studies suffer from methodological issues, such as lack of blinding, heterogeneity, and lack of long-term follow-up.(95) **(EG 2)** A review article states that GnRH agonists can be used as monotherapy for treatment of abnormal uterine bleeding in patients who have failed other medical therapies and are not candidates for or refuse surgical treatments, including hysterectomy. GnRH agonists can also be used to treat abnormal uterine bleeding before hysterectomy or before fertility treatment. GnRH agonists are associated with loss of bone mineral density if used for more than 6 months.(96) **(EG 2)**

For endometriosis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review and meta-analysis, and other review articles and guidelines, indicate effectiveness of GnRH agonists in relieving symptoms such as pelvic pain, dysmenorrhea, dyspareunia, pelvic tenderness, and induration in patients who fail first-line therapy with NSAIDs and oral contraceptives.(26)(99)(100)(102)(103) **(EG 1)** A systematic review of randomized trials reported that leuprolide was equally effective for treatment of endometriosis as compared with continuous oral contraceptives.(101) **(EG 1)** Treatment with GnRH agonists is associated with hypoestrogenic side effects, including accelerated bone loss. The use of add-back therapy, such as low-dose combined oral contraceptive pills, progestins only, or bisphosphonates, may reduce these adverse effects, and allows the use of GnRH agonists for longer than 6 months.(104) **(EG 2)**

For gender incongruence, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** GnRH agonists may be administered at the time of onset of puberty in youths who are transgender or gender diverse in order to delay further development of secondary sexual characteristics of the individual's sex assigned at birth.(110)(113)(114) **(EG 2)** Puberty suppression reversibly suppresses development of secondary sexual characteristics that are not consistent with an individual's experienced gender, which may improve gender dysphoria and allow additional time for the individual to live in the experienced gender before making a decision about any additional irreversible therapy (eg, surgery).(109)(113)(115)(116) **(EG 2)** GnRH agonists may be used as part of gender-affirming hormone therapy in postpubertal patients (eg, for suppression of uterine bleeding in transgender males and as antiandrogen therapy in transgender females).(109)(117)(118)(119) **(EG 2)** A prospective observational study of 36 transgender and gender-diverse adolescents (average age 14.2 years) treated with psychological assessment followed by triptorelin found, at 3-month to 12-month follow-up, improvement from baseline in psychological functioning, including decreased suicidality, depression, anxiety, and body uneasiness; these psychological effects were

correlated with changes in sex steroid levels and other physical changes (eg, height, body mass index, waist-hip ratio, acne) seen with treatment.(120) **(EG 2)**

For uterine leiomyomas (fibroids), evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis and systematic review states that GnRH agonists are effective in reducing uterine and leiomyoma volume, increasing preoperative hemoglobin level, and reducing intraoperative blood loss, operation time, and complication rates.(122) **(EG 1)** A comparative effectiveness review concludes that, while GnRH agonists reduce the size of leiomyomas and the overall size of the uterus and decrease bleeding and leiomyoma-related pain, there is a lack of data on long-term follow-up after treatment discontinuation.(123) **(EG 1)** A practice bulletin suggests effectiveness of GnRH agonists in temporarily alleviating symptoms, such as abnormal uterine bleeding, due to the presence and size of the leiomyomas.(121) **(EG 2)** Case series suggest effectiveness of GnRH agonists in reducing leiomyoma size prior to laparoscopic myomectomy.(124)(125) **(EG 2)** A systematic review and meta-analysis of 23 studies found that administration of GnRH agonists prior to myomectomy was significantly associated with decreased blood loss and need for transfusion, and possibly less significantly associated with decreased formation of uterine adhesions.(126) **(EG 1)**

Inconclusive or Non-Supportive Evidence

For chronic pelvic pain in women, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review and meta-analysis found only a single study with 47 women with chronic pelvic pain (not due to endometriosis, primary dysmenorrhea, acute pelvic inflammatory disease, or irritable bowel syndrome), which suggested that pelvic pain scores were incrementally improved after 1 year of treatment with goserelin as compared with those who took medroxyprogesterone acetate; however, the authors indicated that confirmatory studies are required.(4) **(EG 1)**

For idiopathic short stature, evidence demonstrates a lack of net benefit; additional research is recommended. **(RG C1)** An international consensus statement and review articles conclude that GnRH agonists show a variable effect on adult height gain and are generally not recommended.(5)(6)(7)(8) **(EG 2)**

For male infertility, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** Case series suggest that pulsatile use of GnRH agents may increase testicular volume(9) and induce fertility in males with hypogonadotropic hypogonadism, but such drugs are currently not approved for this indication.(10)(11) **(EG 2)** Pulsatile administration of GnRH agonist requires portable pumps and presents no incremental advantage as compared with gonadotropin therapy.(12) **(EG 2)** A randomized trial of 220 men with hypogonadotropic hypogonadism compared treatment with either pulsatile GnRH or the combination of human chorionic gonadotropin plus human menopausal gonadotropin (HCG/HMG) and found, at 6-month follow-up, that GnRH was associated with higher rates of spermatogenesis as compared with HCG/HMG. However, the authors noted that heterogeneity of prior endocrine treatments and genetic variance among included patients may have limited the results, and further studies were recommended.(13) **(EG 1)**

For menorrhagia prevention in premenopausal women with anticipated chemotherapy-induced thrombocytopenia, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A review found limited evidence on the role of leuprolide for menorrhagia prevention in premenopausal women with thrombocytopenia but noted that more research with large-scale trials is warranted.(14) **(EG 2)** While a practice guideline endorsed the use of leuprolide in this setting, it acknowledged that available evidence includes only uncontrolled cohort studies or case control studies.(15) **(EG 2)**

For undescended testes, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review of 14 studies addressing various types of hormonal therapy for undescended testes found that, while testicular descent occurred in some actively treated patients, studies were generally of poor quality, and descent rates seldom exceeded those of patients treated with placebo by more than 10%.(16) **(EG 1)** A systematic review and meta-analysis of 10 case control or randomized controlled studies found that, while some boys with undescended testes may benefit from an adjunctive GnRH agonist with orchidopexy in terms of improved fertility index, further studies are necessary to determine and confirm which patients are the optimal candidates for treatment.(17) **(EG 1)** A review article found that GnRH agonists have been investigated as an alternative to human chorionic gonadotropin, but results are both incomplete and conflicting, and do not suggest high efficacy. Surgery is still considered the preferred treatment for this condition.(18) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(127); 42 CFR 422.101(128)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(129); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(130); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(131)

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Footnotes

[A] For precocious puberty, nafarelin acetate is administered as an intranasal spray in each nostril and continued until resumption of puberty is desired.(2) Leuprolide may be administered every 1 or 3 months as an intramuscular injection, with careful subsequent monitoring of hormonal levels, bone age, and Tanner staging to confirm downregulation.(62)(68)(69) Leuprolide may be administered every 6 months as a subcutaneous injection, with careful monitoring of hormonal levels and height.(62)(70) Histrelin is administered as a subcutaneous implant of a small capsule in the upper arm every 12 months.(71)(72)(73) Triptorelin is administered every 24 weeks as an intramuscular injection, with careful subsequent monitoring of hormonal levels, bone age, and height to confirm downregulation.(74) (75)(76) [A in Context Link 1]

[B] Bone age is determined by comparison of a left hand and wrist x-ray with a standardized reference.(87)(88) [B in Context Link 1]

[C] There is no consensus about when to stop therapy for central precocious puberty; discontinuation is dependent on a number of patient-specific characteristics as well as family preference.(90)(91)(93)(94) [C in Context Link 1]

[D] For dysfunctional uterine bleeding, goserelin acetate may be administered as a subcutaneous implant for endometrial thinning prior to endometrial ablation.(1) [D in Context Link 1]

[E] For endometriosis, goserelin acetate may be administered as a subcutaneous implant.(1) The recommended duration of therapy for both goserelin acetate implants and intranasal nafarelin acetate is 6 months; retreatment is not recommended.(1)(2) If symptoms recur, only one additional 6-month course of leuprolide (with concomitant norethindrone acetate unless contraindicated) may be given.(97)(98) [E in Context Link 1]

[F] For gender incongruence in peripubertal individuals, a specialty society guideline recommends initiation of a GnRH agonist from the time of starting puberty until the individual elects to initiate additional gender-affirming hormone therapy, if desired, after the individual has demonstrated persistent gender incongruence and is able to provide informed consent, typically at approximately age 16 years. (109) [F in Context Link 1]

[G] Gender incongruence is a difference between an individual's experienced gender and the gender expected of them based on sex assigned at birth. Gender-affirming medical and surgical treatments, including puberty suppression, hormone therapy, surgery, or a combination, may play a role in supporting the development and expression of a person's experienced gender identity by allowing them to live and be accepted as the experienced gender. Gender diversity is not pathologic, and gender incongruence is not defined by the presence of distress. However, gender incongruence may be associated with clinically significant distress and impairment (ie, gender dysphoria), and gender-affirming medical therapy may reduce the likelihood of gender dysphoria. Individuals seeking gender-affirming interventions should be evaluated by a healthcare professional with experience caring for transgender and gender-diverse individuals and the ability to identify coexisting mental health or psychosocial conditions and conditions that can be mistaken for gender incongruence. Individuals seeking gender-affirming medical interventions should have demonstrated a marked and persistent gender incongruence, which includes incongruence that has not previously been disclosed to others.(110) [G in Context Link 1, 2]

[H] For uterine leiomyomas, leuprolide may be administered in depot form prior to operative intervention.(97)(98) [H in Context Link 1]

Codes

HCPCS: C9399, J1620, J1675, J1950, J1951, J1952, J1954, J3315, J3316, J3490, J9202, J9217, J9218, J9219, J9225, J9226, S9560 [Hide]

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